

A Prospective Randomized Study of Anterior Cruciate Ligament Reconstruction With Autograft, γ -Irradiated Allograft, and Hybrid Graft

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Purpose: To compare the results of patients who underwent anterior cruciate ligament (ACL) reconstruction with autograft, γ -irradiated allograft, or hybrid graft in a prospective randomized study with a minimum clinical follow-up period of 5 years. **Methods:** In this prospective, randomized, comparative study, 102 patients with ACL tears underwent ACL reconstruction with autograft (gracilis and semitendinosus tendons), γ -irradiated allograft (tibialis anterior tendons), or hybrid graft (γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft). Laboratory testing of the erythrocyte sedimentation rate and C-reaction protein level were performed; clinical results were evaluated with the KT-1000 arthrometer (MEDmetric, San Diego, CA), Lachman test, Lysholm score, Tegner activity score, and International Knee Documentation Committee evaluation (both objective and subjective). **Results:** There were 32 patients in the autograft group, 31 in the hybrid graft group, and 32 in the γ -irradiated allograft group at last follow-up. No differences were found among the 3 groups regarding patient demographic data and the duration from injury to operation ($P > .05$). The C-reaction protein and erythrocyte sedimentation rate values were statistically higher in the γ -irradiated allograft group than in the other 2 groups on the third, seventh, and fourteenth days ($P < .05$). No significant differences were found between the autograft and hybrid graft groups ($P > .05$). The KT-1000 examination showed more anterior laxity in the γ -irradiated allograft group than in the other 2 groups ($P < .05$). No significant differences in the Lachman test and pivot-shift test findings were found among the 3 groups ($P > .05$). The Lysholm score, Tegner activity score, and International Knee Documentation Committee evaluation (subjective and objective) showed no differences among the 3 groups ($P > .05$). **Conclusions:** Patients undergoing primary ACL reconstruction with hybrid graft or autograft had satisfactory and similar objective and subjective clinical results. On KT-1000 measurement of anteroposterior knee laxity, both the autograft and hybrid graft groups showed statistically significant differences compared with the γ -irradiated allograft group. **Level of Evidence:** Level II, prospective comparative study.

Anterior cruciate ligament (ACL) reconstruction with bone–patellar tendon–bone (BPTB) autograft has been considered a common procedure and the gold standard over the past few decades. Combined semitendinosus tendon–gracilis tendon autografts have become increasingly popular, with several reports showing that the results are comparable in terms of

patient satisfaction, knee laxity measurements, and functional tests.¹⁻³ The major benefits of ACL reconstruction with autograft are earlier incorporation and the absence of a host immune response against the donor tissue or disease transmission. However, there remains a considerable amount of controversy over the use of allograft for ACL reconstruction as a reasonable alternative to autograft.^{1,4-6} There are several advantages to the use of allograft, such as reduced operating time, elimination of donor-site morbidity, smaller incisions, and availability of larger grafts. Furthermore, allografts might significantly change the initial biomechanical properties after γ -irradiation; whether the changes in biomechanical properties affect the clinical outcomes of ACL reconstruction with γ -irradiated allograft still remains controversial.⁷⁻¹²

The erythrocyte sedimentation rate (ESR) and C-reaction protein (CRP) level are the most useful routine

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laboratory screening tests for diagnosing potential infection after surgery and determining the response to treatment. Previous studies have found similar clinical results between allograft and autograft.¹⁰⁻¹² The purpose of this study was to compare the results of patients who underwent ACL reconstruction with autograft, γ -irradiated allograft, or hybrid graft in a prospective randomized study with a minimum clinical follow-up period of 5 years.

Methods

This research was designed as a prospective, randomized controlled trial of ACL reconstruction with autograft compared with γ -irradiated allograft or hybrid graft, which was based on an intent-to-treat protocol and analysis. This study was approved by the Institutional Review Board of The Third Hospital of Hebei Medical University. Signed informed consent was obtained from each patient. The clinical investigations were conducted following the principles expressed in the Declaration of Helsinki. Patients provided informed consent to participate in the study; those who did not wish to participate in this study were not enrolled.

The inclusion criterion was a unilateral ACL rupture verified clinically by positive Lachman test and positive pivot-shift test findings. All patients had undergone a preoperative magnetic resonance imaging scan to confirm the ACL rupture. All patients with normal preoperative CRP (<10 mg/L) and ESR (≤ 15 mm/h) values were included. The exclusion criteria were combined multiple-ligament injuries, previous ACL surgery, contralateral knee ligament injury, and radiographically verified osteoarthritis. Patients with pre-existing metabolic pathologies such as diabetes mellitus or uremia were excluded from the study. Patients who could not

finish the minimum clinical follow-up period of 5 years were also excluded.

Before the study, the sample size was justified based on the change in anteroposterior laxity from preoperatively to the final postoperative follow-up. We determined that a sample size of 26 patients was required (power of 0.8 and $P < .05$). Therefore we planned to enroll 34 patients per group to account for patients lost to follow-up. After signing informed consent forms, the patients were randomly assigned to 1 of 3 groups (Fig 1).

Surgical Techniques

Regarding graft preparation, on the day of the operation, the autografts (gracilis and semitendinosus tendons) were harvested through a standard oblique incision over the pes anserinus by a tendon stripper and were cleaned of soft tissue on a graft preparation station board. Both tendons were sutured together, and a 4-strand graft was made. All the allografts were irradiated and supplied by a certified tissue bank, which had policies for serologic and microbiologic testing in accordance with guidelines set forth by the American Association of Tissue Banks and the Food and Drug Administration. The irradiated allografts received an irradiation dose of 2.5 Mrad before distribution. The γ -irradiated allografts (tibialis anterior tendons) were thawed in sterile physiological fluid at room temperature before preparation and were sutured together to form a 4-strand graft. The gracilis and semitendinosus tendons were identified and retracted with hook retractors; then, only the semitendinosus tendon was harvested with an open-loop tendon stripper. If the harvested semitendinosus tendon is too small, it can be augmented with tibialis anterior tendon allograft. The hybrid graft was composed of the tibialis anterior tendon (γ -irradiated allograft) and semitendinosus tendon

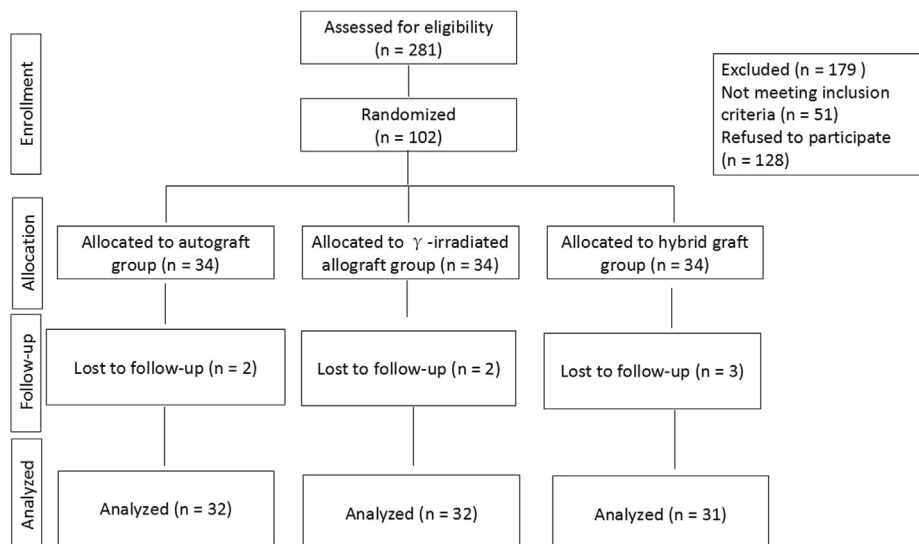


Fig 1. CONSORT (Consolidated Standards of Reporting Trials) flow diagram.

(autograft) together to create a soft-tissue graft suitable for ACL reconstruction. The minimum diameter of the graft was 8 mm.

Before the reconstruction, irreparable tears of the meniscus were dealt with by partial meniscectomy. Tears in the red-red or red-white zone were repaired by an inside-out technique. The same surgical technique was used in all the groups. All patients underwent an examination under anesthesia and diagnostic arthroscopy to confirm the preoperative diagnosis. First, the remnant fibers of the ACL were debrided to find the anatomic attachments on the tibia and femur. After the tibial guide pin was placed at the footprint of the ACL adjacent to the anterior horn of the lateral meniscus, the tibial tunnel was reamed. After the anatomic femoral insertion was identified, the femoral tunnel was created independently of the tibial tunnel based on the native ACL insertional anatomy on the femoral side through the anteromedial-portal approach, from inside to outside. The minimum diameter of the graft was 8 mm. The femoral guide pin was placed at the 10-o'clock (right leg) or 2-o'clock (left leg) position, 5 mm anteriorly from the posterior bony edge of the intercondylar wall, which was observed arthroscopically from the anteromedial portal, with the knee flexed 120°. All tunnels were reamed according to graft size determined by a graft sizer. A probe was inserted to help pass the graft into the femoral tunnel. The femoral tunnel was fixed with an EndoButton (Smith & Nephew Endoscopy, Andover, MA) when the graft was put in place. The EndoButton was flipped and fixed on the femoral cortical surface, avoiding possible secondary slackening. The tibial tunnel was fixed with Bio-Composite interference screws (Arthrex, Naples, FL) at 20° of knee flexion under manual maximum tension, and these screws were placed in the most distal part of the bone tunnel that was in contact with cortical bone. During graft fixation, the tibia was loaded with a maximum posterior force to minimize graft laxity present at the time of surgery.

Rehabilitation

All patients followed the same ACL rehabilitation program. Patients began performing closed-chain exercises with a hinged brace immediately postoperatively. Range-of-motion (ROM) exercises were started 2 days after the operation, with nearly normal ROM achieved in 4 weeks. Patients achieved normal ROM and were allowed full weight bearing at 12 weeks. Functional activities including walking and jogging were allowed at 6 months postoperatively. Patients were permitted to return to work or full sports activity 12 months after the operation.

Clinical Evaluation

CRP and ESR values were evaluated before surgery and on days 1, 3, 7, 14, 21, and 24 postoperatively. The immunoturbidimetric method was used to test the level of serum CRP (<10 mg/L). The modified Westergren method was used to measure the ESR (≤ 15 mm/h). Examination of knee laxity included the Lachman test and pivot-shift test. Instrumented anteroposterior laxity of the affected side and side-to-side differences were measured with the KT-1000 arthrometer (MEDmetric, San Diego, CA). The Lysholm score, Tegner activity score, and International Knee Documentation Committee (IKDC) evaluation (both objective and subjective) were completed preoperatively and at each follow-up evaluation. When the follow-up period was longer than 5 years, the most recent data available were used for statistical analysis. Clinical evaluation assessments were performed before the operation and at last follow-up. All of the examinations were performed by the senior physicians (D.S., X.Y.) twice at each time point, neither of whom was the operating surgeon and both of whom were blinded to the clinical findings.

Statistical Analysis

We used SPSS software for Windows (version 17.0; SPSS, Chicago, IL) for statistical analysis. The power analysis indicated that a 2.0-mm difference in

Table 1. Demographic Characteristics

	Autograft	γ -Irradiated Allograft	Hybrid Graft	<i>P</i> Value
Age, mean $\pm$ SD, yr	29.8 $\pm$ 7.9	30.5 $\pm$ 6.1	31.6 $\pm$ 8.2	.128
Sex, male/female, n	17/15	15/17	18/13	.672
Injury-to-operation interval, mean $\pm$ SD, mo	2.2 $\pm$ 1.3	2.5 $\pm$ 1.8	2.1 $\pm$ 1.1	.739
Type of injury, n				
Sports injury	25	24	26	
Fall	4	5	3	
Traffic accident	3	3	2	
Associated injury, n				
Medial meniscus injury	17	17	17	
Lateral meniscus injury	15	15	14	
Follow-up, mean $\pm$ SD (range), yr	5.8 $\pm$ 0.9 (5-7)	6.1 $\pm$ 0.3 (5-7)	5.9 $\pm$ 0.6 (5-7)	.172

NOTE. The autograft group received gracilis and semitendinosus tendons, the hybrid graft group received γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft, and the γ -irradiated allograft graft received tibialis anterior tendons.

Table 2. Treatment Methods for Meniscal Tears

Treatment Method	Autograft	Hybrid Graft	γ -Irradiated Allograft	<i>P</i> Value
None	6	7	3	.695
Repair	2	2	3	
Partial meniscectomy	24	22	26	

NOTE. Data are presented as number of patients unless otherwise indicated. The autograft group received gracilis and semitendinosus tendons, the hybrid graft group received γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft, and the γ -irradiated allograft graft received tibialis anterior tendons.

anteroposterior laxity from preoperatively to final follow-up among the different groups, as well as a corresponding SD of 2.0 mm, should be considered clinically relevant.¹³ We performed repeated-measures analysis of the ESR and CRP values and used the Student-Newman-Keuls test for group-to-group comparisons. Comparison between the 3 different groups was conducted by use of 1-way analysis of variance with post hoc testing by the Student-Newman-Keuls test. The χ^2 test was used to compare categorical variables among the groups. The paired *t* test was used to compare the preoperative and postoperative differences in results. In all analyses, significance was defined as $P < .05$. Results are presented as mean $\pm$ standard deviation.

Results

From July 2005 to June 2008, 281 patients with ACL ruptures underwent ACL reconstruction at our institution; 102 of these patients were eligible to participate in the study and had preoperative CRP and ESR values available. There were 128 patients who did not wish to participate in this study. Fifty-one patients were excluded because of combined multiple-ligament injuries, previous injury, or previous surgery on the affected knee. During the 5-year minimum follow-up period (mean, 5.9 ± 0.6 years; range, 5 to 7 years), there were 32 patients in the autograft group, 31 in the hybrid graft group, and 32 in the γ -irradiated allograft

group. Five patients had changed their telephone numbers and could not be contacted, and 2 patients moved far from our hospital. The rest of the patients were followed up for a minimum of 5 years. There were no statistical differences in sex, age, injury-to-operation interval, or follow-up among the 3 groups ($P > .05$) (Table 1). There were no cases of major neurovascular, infectious, vascular, or wound complications. Treatments performed for tears of the meniscus are shown in Table 2. No patient needed additional surgery because of recurrent or residual symptoms. The mean operating time in the autograft group (75.3 ± 12.8 minutes) was significantly longer than that in the allograft group (58.9 ± 10.2 minutes, $P < .05$). The mean operating time in the hybrid graft group (62.9 ± 15.3 minutes) was slightly longer than that in the allograft group, but the difference was not significant ($P > .05$).

The CRP and ESR values increased in all 3 groups of patients. The CRP values increased on the first postoperative day; the peak values were then reached on the third postoperative day. The ESR values also increased on the first postoperative day, but they reached their peak on the seventh day postoperatively. The CRP and ESR values decreased after reaching their peak levels. The CRP values returned to nearly normal levels on the fourteenth day postoperatively. The CRP and ESR values were statistically higher in the γ -irradiated allograft group than in the autograft and hybrid graft groups on the third, seventh, and fourteenth days ($P < .05$). However, when we compared the CRP and ESR values after ACL reconstruction in the hybrid graft and autograft groups, no significant differences were found between the 2 groups ($P > .05$) (Table 3).

The Lachman test and pivot-shift test findings improved significantly from preoperatively to last follow-up; however, no significant difference was found among the 3 groups ($P > .05$) (Table 4). On KT-1000 measurement of anteroposterior knee laxity, the hybrid graft group showed slightly superior stability, but no statistically significant difference was found

Table 3. CRP and ESR Values Before Surgery Through Postoperative Day 21

	Preoperative	Postoperative				
		Day 1	Day 3	Day 7	Day 14	Day 21
CRP level, mg/L						
Autograft	1.3 ± 0.8	18.3 ± 5.5	53.8 ± 9.3	22.8 ± 6.3	5.6 ± 3.3	2.3 ± 0.9
Hybrid graft	1.6 ± 1.1	20.5 ± 3.7	55.2 ± 11.5	25.3 ± 4.6	7.3 ± 1.1	3.6 ± 1.5
γ-Irradiated allograft	1.8 ± 0.9	23.3 ± 6.6	60.3 ± 9.3	29.1 ± 7.2	9.3 ± 1.1	5.3 ± 2.1
ESR, mm/h						
Autograft	8.2 ± 1.9	18.5 ± 5.3	32.7 ± 8.2	49.6 ± 9.3	23.6 ± 5.3	11.5 ± 6.9
Hybrid graft	8.5 ± 2.8	20.2 ± 6.9	35.3 ± 7.3	52.1 ± 8.6	28.5 ± 6.7	13.3 ± 5.2
γ-Irradiated allograft	8.8 ± 3.3	25.1 ± 7.3	39.7 ± 9.5	56.6 ± 10.2	33.2 ± 7.2	15.1 ± 3.2

NOTE. Data are presented as mean $\pm$ standard deviation. The autograft group received gracilis and semitendinosus tendons, the hybrid graft group received γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft, and the γ -irradiated allograft graft received tibialis anterior tendons.

CRP, C-reactive protein; ESR, erythrocyte sedimentation rate.

Table 4. Comparison of Knee Functional Assessment and Stability at Final Follow-up

	Autograft	Hybrid Graft	γ -Irradiated Allograft	<i>P</i> Value
Overall IKDC scor				.880
Normal	9	9	8	
Nearly normal	20	19	21	
Abnormal	3	3	2	
Severely abnormal	0	0	1	
Pivot-shift test				.249
Grade 0	28	28	23	
Grade I	4	3	8	
Grade II	0	0	1	
Grade III	0	0	0	
Lachman test				.209
Grade 0	26	27	19	
Grade I	4	3	9	
Grade II	2	1	3	
Grade III	0	0	1	

NOTE. Data are presented as number of patients unless otherwise indicated. The autograft group received gracilis and semitendinosus tendons, the hybrid graft group received γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft, and the γ -irradiated allograft graft received tibialis anterior tendons.

IKDC, International Knee Documentation Committee.

compared with the autograft group ($P > .05$). Both the autograft and hybrid graft groups showed statistically significant differences when compared with the γ -irradiated allograft group ($P < .05$) (Table 5).

There were no significant differences in the Lysholm score, Tegner activity score, or IKDC evaluation (subjective and objective) among the 3 groups preoperatively, but there were significant improvements at last follow-up compared with preoperative values in the 3 groups, as shown in Tables 4 and 6 ($P < .05$). At last follow-up, on the basis of objective IKDC scores, 29 patients (90.6%) in the autograft group, 28 (90.3%) in the hybrid graft group, and 26 (87.5%) in the allograft group had an IKDC knee classification of normal or nearly normal; the 3 groups did not show any significant difference ($P > .05$). Comparison of the objective IKDC scores showed no significant difference among the 3 groups at last follow-up. The mean subjective IKDC scores were 87.5 ± 3.2 , 89.8 ± 5.7 , and 83.8 ± 6.9 for the autograft group, hybrid graft group, and γ -irradiated allograft group, respectively, as shown in Table 6 ($P > .05$).

Discussion

Our results showed that the postoperative KT-1000 measurement of anteroposterior knee laxity was significantly superior in the autograft and hybrid graft groups when compared with the allograft group. Concerning the results of the IKDC evaluation, no statistically significant differences could be found between the patients in the autograft group and those in the hybrid graft group. However, the results for patients in both the autograft and hybrid graft groups were significantly superior to those for patients in the γ -irradiated allograft group. There were no significant differences in the Tegner activity score or Lysholm knee score among the 3 groups. Moreover, no significant differences were found between the autograft and hybrid graft groups. This finding might suggest that patients with higher activity levels could choose not only autograft but also hybrid graft during ACL procedures. In the future, we would use a robust set of subjective and objective outcome scores.

According to anteroposterior laxity measurements, as well as Lachman test and pivot-shift test findings, Mehta et al.¹⁴ reported a higher rate of revision ACL reconstruction associated with the use of γ -irradiated allograft during the primary ACL reconstruction. This higher revision rate might have been associated with graft irradiation. Moreover, Guo et al.¹⁵ reported statistically poorer KT-1000 results and a higher failure rate in the γ -irradiated allograft group compared with the autograft group. They found that the high revision rate was related only to the use of γ -irradiated allograft, which means that γ -irradiation might weaken the graft. Such findings might suggest that γ -irradiated allograft is not a good graft candidate for ACL reconstruction. Similarly, Sun et al.¹⁶ reported excellent outcomes regarding knee functional assessment after ACL reconstruction with irradiated hamstring tendon allograft but relatively poor anteroposterior laxity compared with autograft. On the contrary, Rihn et al.¹⁷ reported that patients undergoing ACL reconstruction with irradiated BPTB allograft had similar clinical outcomes to patients undergoing reconstruction with BPTB autograft. Sun et al.¹⁸ compared the clinical outcomes of arthroscopic ACL reconstruction with autograft and irradiated allograft. Patients in the autograft group were satisfied with their clinical

Table 5. Comparison of Knee Laxity at Final Follow-up According to KT-1000 Testing

	Preoperative		Postoperative	
	Mean $\pm$ SD, mm	Confidence Interval, mm	Mean $\pm$ SD, mm	95% Confidence Interval, mm
Autograft	8.4 ± 2.0	7.7-9.1	2.1 ± 1.6	1.6-2.6
Hybrid graft	8.5 ± 3.2	7.3-9.6	2.0 ± 1.5	1.4-2.6
γ -Irradiated allograft	8.7 ± 2.7	7.7-9.7	3.5 ± 1.2	3.1-4.1
<i>P</i> value	.893		.025	

NOTE. The autograft group received gracilis and semitendinosus tendons, the hybrid graft group received γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft, and the γ -irradiated allograft graft received tibialis anterior tendons.

Table 6. Subjective Evaluation and Activity Level Scores After Surgery

	Autograft	Hybrid Graft	γ -Irradiated Allograft	P Value
Tegner activity score	7.3 $\pm$ 1.3	7.5 $\pm$ 1.5	7.0 $\pm$ 2.1	.416
Lysholm score	91.3 $\pm$ 11.5	90.5 $\pm$ 10.2	88.7 $\pm$ 8.6	.213
Subjective IKDC score	87.5 $\pm$ 3.2	89.8 $\pm$ 5.7	83.8 $\pm$ 6.9	.353

NOTE. Data are presented as mean $\pm$ standard deviation unless otherwise indicated. The autograft group received gracilis and semitendinosus tendons, the hybrid graft group received γ -irradiated tibialis anterior tendon allograft and semitendinosus tendon autograft, and the γ -irradiated allograft graft received tibialis anterior tendons.

IKDC, International Knee Documentation Committee.

outcomes, whereas the difference in instrumented laxity between the 2 groups was significant and no significant difference in functional test results was found.

At our hospital, all the allografts were irradiated and supplied by a certified tissue bank that had policies for serologic and microbiologic testing in accordance with guidelines set forth by the American Association of Tissue Banks and the Food and Drug Administration. The irradiated allografts received an irradiation dose of 2.5 Mrad before distribution. On KT-1000 measurement of anteroposterior knee laxity, the hybrid graft group showed slightly better stability, but no statistically significant differences were found compared with the autograft group. Both the autograft and hybrid graft groups showed statistically significant differences when compared with the γ -irradiated allograft group. Our results suggest that γ -irradiation could decrease the tension of allograft. However, when comparing the clinical outcomes of ACL reconstruction with hybrid graft versus autograft, we found no significant difference between the 2 groups according to the subjective and objective clinical results.

CRP and ESR values are valuable laboratory tests for detecting postoperative infection. The CRP and ESR values increased dramatically when infection, inflammation, and injury occurred. The combination of the CRP and ESR values was used as a diagnosis marker of infection after surgery and for determination of the response to treatment. This study found that the CRP levels were higher early after surgery and then returned to normal after approximately 20 days. On the contrary, the ESR values increased in the first week after surgery and had not normalized by 20 days postoperatively. Data collected in all of the groups showed that the ESR values increased after surgery, peaked between the third and seventh days, and then dropped gradually. The faster increase and decrease in CRP, in comparison with ESR, to normal levels indicate that the CRP level is a more sensitive indicator of postoperative complications. Furthermore, the trends in CRP and ESR values were

similar in patients in the autograft, γ -irradiated allograft, and hybrid graft groups. No patient had a postoperative infection develop. The CRP and ESR values were statistically higher in the γ -irradiated allograft group than in the autograft and hybrid graft groups. However, when comparing the clinical outcomes of ACL reconstruction with hybrid graft versus autograft, we found no significant differences between the 2 groups according to the CRP and ESR values.

Limitations

This study had several limitations. This is a single-surgeon study, and the results may not be generalized. At the clinical follow-up, all patients were examined by an independent examiner who was not the operating surgeon and who did not know which type of graft was used for ACL reconstruction. The patients, however, were aware of the type of procedure at the follow-up examination because they had been informed by the senior surgeon. The study groups were of limited size, which may influence the validity of the conclusions, and a longer double-blind follow-up period is required to further evaluate the clinical outcomes of γ -irradiated and hybrid grafts. Although the same standardized therapy protocol was followed by all patients after surgery, physical therapy provided outside of our hospital might also have affected the clinical outcomes.

Conclusions

Patients undergoing primary ACL reconstruction with hybrid graft or autograft had satisfactory and similar objective and subjective clinical results. On KT-1000 measurement of anteroposterior knee laxity, both the autograft and hybrid graft groups showed statistically significant differences compared with the γ -irradiated allograft group.

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